Matthias Brandt

Senior Software Developer — Backend and Distributed Systems

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PROFESSIONAL SUMMARY

Senior backend engineer with 13 years of experience designing distributed systems for logistics, payments, and automotive IoT platforms. Specializes in event-driven architecture with Apache Kafka, Go, and Java services that must stay correct under high throughput and partial failure. Has led architecture decisions on idempotent processing, the transactional outbox pattern, and saga-based workflows, and has reduced API latency and production incidents in systems handling tens of thousands of events per second. Works closely with product managers and site reliability engineers to turn service-level objectives into concrete design constraints. Mentors mid-level engineers, runs system design reviews, and has a strong interest in consensus protocols, observability, and testable distributed designs.

TECHNICAL SKILLS

Programming languages: Advanced: Go, Java, SQL. Working knowledge: Kotlin, Python, TypeScript, Bash.

Frameworks and libraries: Spring Boot, gRPC, Protocol Buffers, Pact, Testcontainers, JUnit 5.

Architecture and system design: Event-driven architecture, microservices, CQRS, saga orchestration, transactional outbox, idempotent consumers, distributed rate limiting, REST and gRPC API design, architecture decision records (ADRs).

Messaging and databases: Advanced: Apache Kafka, Kafka Streams, PostgreSQL, Redis. Working knowledge: Apache Cassandra, ClickHouse, RabbitMQ.

Cloud, DevOps, and observability: AWS (EKS, MSK, RDS, S3), Kubernetes, Helm, Terraform, GitHub Actions, Prometheus, Grafana, OpenTelemetry, Jaeger.

Industry experience: Logistics and shipment tracking, payments and ledgers, automotive telematics and IoT.

Leadership and collaboration: Technical leadership, mentoring, system design reviews, incident command, on-call process design, collaboration with product managers and SREs.

PROFESSIONAL EXPERIENCE

Kestrel Logistics Cloud (logistics SaaS) — Berlin, Germany

Senior Software Engineer, Distributed Systems — *Mar 2021 – Present*

- Led the redesign of shipment event ingestion from a single PostgreSQL write path to an event-driven architecture on Apache Kafka and Go services, sustaining 45,000 events per second at peak with p99 end-to-end latency under 180 ms.
- Wrote the architecture decision records for idempotent consumers and the transactional outbox pattern, eliminating duplicate shipment status updates that had caused about 1,200 customer support tickets per quarter.
- Reduced p99 latency of the public tracking API from 420 ms to 160 ms with Redis read models, connection pooling, and query plan fixes, after agreeing latency SLOs with the product manager.
- Introduced consumer-driven contract tests with Pact and property-based tests, raising test coverage of core routing services from 58% to 84% and halving production rollbacks over three release cycles.
- Mentor four mid-level engineers and run the fortnightly system design review for the 30-person backend organization.

Lumenpay GmbH (payments) — Berlin, Germany

Senior Backend Engineer — *Jun 2017 – Feb 2021*

- Designed a double-entry ledger service in Java and Spring Boot on PostgreSQL, processing 2.3 million transactions per day with no reconciliation breaks in the 18 months after launch.
- Replaced distributed transactions in the refund flow with saga orchestration across five services, lowering the refund failure rate from 0.9% to 0.1%.
- Led the migration of 22 services from virtual machines to Kubernetes with Helm charts and GitHub Actions, moving deployment frequency from weekly releases to several deployments per day.
- Built an immutable audit event log with the compliance team, allowing audit evidence requests to be answered within one day instead of one week.

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Northwind Telematics (automotive IoT) Munich, Germany
Software Engineer *Sep 2013 – May 2017*

- Built a gRPC telemetry gateway in Go that handled data from about 120,000 connected vehicles with automatic backpressure during network reconnect storms.
- Implemented time-series storage on Apache Cassandra with time-window partitioning and TTLs, reducing storage cost by 35% while keeping 13 months of history.
- Wrote on-call runbooks for the gateway and ingestion services, helping cut mean time to recovery from 70 to 25 minutes.

SELECTED PROJECTS

Ferrule — distributed rate limiter (open source)
Technologies: Go, Redis, Lua scripting, Prometheus

- **Problem:** Internal APIs used per-instance rate limits, so customers saw inconsistent limits whenever autoscaling changed the number of pods.
- **Contribution:** Designed and implemented a token-bucket limiter backed by atomic Redis Lua scripts, with a local fallback mode when Redis is unavailable and metrics for every decision.
- **Result:** Adopted by three teams at Kestrel Logistics Cloud; enforced limits stayed within 2% of target in load tests at 20,000 requests per second.

Kafka consumer chaos harness
Technologies: Go, Testcontainers, Toxiproxy, Apache Kafka, GitHub Actions

- **Problem:** Rebalancing and broker failures caused rare duplicate processing that unit tests could not reproduce.
- **Contribution:** Built a harness that injects latency, partition leader loss, and consumer crashes during integration tests and checks ordering and idempotency invariants.
- **Result:** Found seven ordering and offset-commit defects before release; now runs nightly for 12 services.

Raft-based configuration store (research prototype)
Technologies: Go, Raft, gRPC, Porcupine

- **Problem:** The team needed to weigh a self-built consistent configuration store against a managed one.
- **Contribution:** Implemented leader election, log replication, snapshots, and partition linearizability tests.
- **Result:** Informed the decision to adopt a managed store; now used as design review teaching material.

EDUCATION

Technical University of Rhenwald Rhenwald, Germany
M.Sc. Computer Science, specialization in Distributed Systems *2011 – 2013*

- Thesis: leader election latency in Raft clusters under asymmetric network partitions (final grade 1.3).
- Relevant coursework: distributed algorithms, database systems, network protocols, formal verification.

Technical University of Rhenwald Rhenwald, Germany
B.Sc. Computer Science *2008 – 2011*

- Relevant coursework: operating systems, compilers, algorithms and data structures.

HACKATHONS, CONFERENCES, AND PROFESSIONAL ACTIVITIES

- **Berlin Distributed Systems Meetup (2024)** — Speaker: “Exactly-once is a promise you have to build”, a talk on idempotency and the outbox pattern based on the shipment ingestion migration.
- **EventStream Europe Conference (2023)** — Workshop co-facilitator on Kafka consumer rebalancing and partition design (about 40 attendees).
- **Hack the Grid Hackathon, Munich (2019)** — Team lead; built a 48-hour prototype API that forecast charging-station load from vehicle telemetry using Go, PostgreSQL, and Python. Finalist among 22 teams.

CERTIFICATIONS AND ADDITIONAL QUALIFICATIONS

Certifications (illustrative): Cloud-Native Application Development Track, Meridian Cloud Academy (2020); Advanced Kafka Operations Course, Northgate Data School (2022).

Mentoring and open source: Mentor in the Kestrel Logistics Cloud mentorship program since 2022; maintainer of the Ferrule rate limiter.

Languages: German (native), English (fluent, professional), Spanish (basic).

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